AI-DRIVEN OUTFIT RECOMMENDATION AND MATCHMAKING SYSTEM

Project ID – R26-IT-053

PROJECT PROPOSAL REPORT

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B.Sc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Nowadays, the fast growth of fashion worldwide means people face too many clothing choices every day creating an era of “Decision Fatigue”. Because of this, picking what to wear takes up much more time than before. Even though online shopping sites suggest outfits, their main goal tends to be selling new items instead of helping users organize clothes they already own. Research into smart wardrobe tools often depends on complicated devices connected through technology networks. Some studies just match weather data with outfit suggestions, but these ideas work poorly in hot, humid places. In countries such as Sri Lanka, humidity levels shape how people dress far more than seasonal changes do. Social events also play a big role there when choosing garments. Most current models fail to reflect those real-life conditions accurately.

A fresh approach emerges here - using artificial intelligence to rethink how outfits are suggested and paired. Instead of relying on basic tags, one core piece builds a Semantic Fashion Knowledge Graph, turning raw data into meaningful connections that support logical decisions and clear explanations. What stands out? This structure handles fashion rules like a guide, moving past shallow matches. Another layer dives into context: an engine works on two levels, blending event-based cues with individual style profiles to balance social fit and self-expression. Settings matter too - a smart module tracks micro-climates and fabric behaviors, adjusting suggestions using real-world physics around moisture and environment. Finally, creativity gets a boost through a visual generator powered by diffusion techniques, imagining original ensembles outside existing wardrobes.

A fresh approach begins here: CNNs built on ResNet pull key details from images. Following that, Graph Neural Networks encode relationships within knowledge graphs. Instead of traditional generation, Stable Diffusion shapes full outfits through learned patterns. Evaluation unfolds not only through numbers - precision, recall, and F1-score matter - but also how users feel about suggestions. Benchmarks stack up against older methods, revealing shifts in performance. What stands out is a digital stylist tuned to cultural context. It walks the line where social setting meets individual taste. Reasons behind each choice come spelled out, clear and logic-driven. Earlier gaps between rules and creativity start closing here.

Key words: Outfit Recommendation, Knowledge Graph, Diffusion Models, Fashion AI, Explainable AI, Deep Learning, Tropical Climate Adaptation

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
GNN	Graph Neural Network
KG	Knowledge Graph
NLP	Natural Language Processing
XAI	Explainable Artificial Intelligence
ResNet	Residual Network
ReLu	Rectified Linear Unit
API	Application Programming Interface
JSON	JavaScript Object Notation
SQL	Structured Query Language
Saas	Software as a Service
B2C	Business to Consumer
REST	Representational State Transfer
CDN	Content Delivery Network
SSL	Secure Sockets Layer
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit

1. INTRODUCTION

1.1 Background

The global fashion industry has experienced unprecedented growth over the past decade, with the e-commerce fashion market projected to reach $1.2 trillion by 2025 [1]. This rapid expansion has fundamentally transformed how consumers discover, purchase, and interact with fashion products. However, this abundance of choice has created a paradoxical phenomenon known as "Decision Fatigue"—a cognitive drain that occurs when individuals face an overwhelming number of options [2]. Research indicates that the average individual spends approximately 17 minutes each morning deciding what to wear, accumulating to over 10 days per year dedicated solely to wardrobe decisions [3].

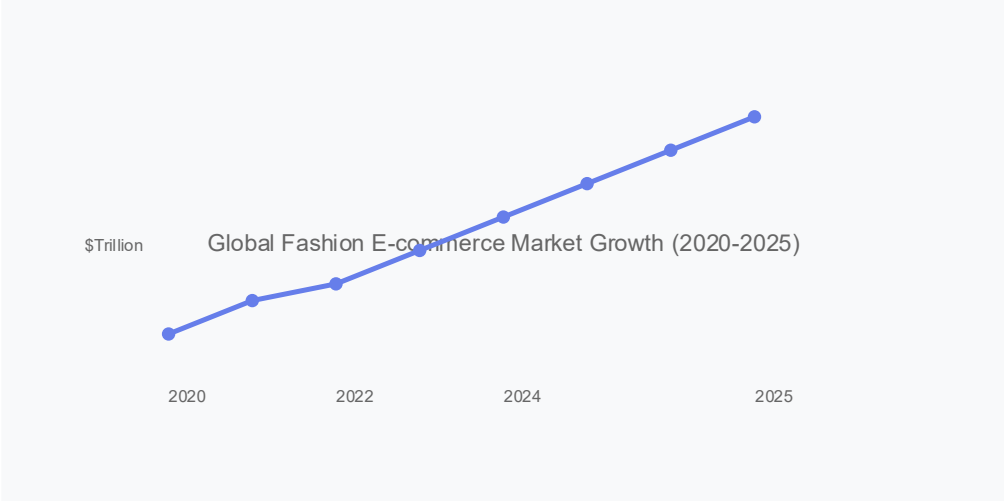


Figure 1. 1 Global Fashion E-commerce Market Growth Projection

Modern e-commerce platforms have developed sophisticated recommendation systems to address this challenge, but these systems are primarily designed to drive new consumption rather than optimizing the use of existing wardrobes. Collaborative filtering and content-based approaches dominate the current landscape, focusing on suggesting new items to purchase based on user behaviour and item similarity [4]. While effective for sales conversion, these approaches fail to address the fundamental problem of wardrobe management and daily outfit selection. The concept of "Smart Wardrobes" has emerged as a potential solution, with existing research exploring various

technological approaches. Early implementations relied heavily on hardware-intensive IoT solutions, utilizing RFID tagging, smart hangers, and sensor-equipped closets to track wardrobe inventory [5]. While technically feasible, these solutions require significant infrastructure investment and lack the intelligence to provide meaningful styling recommendations.

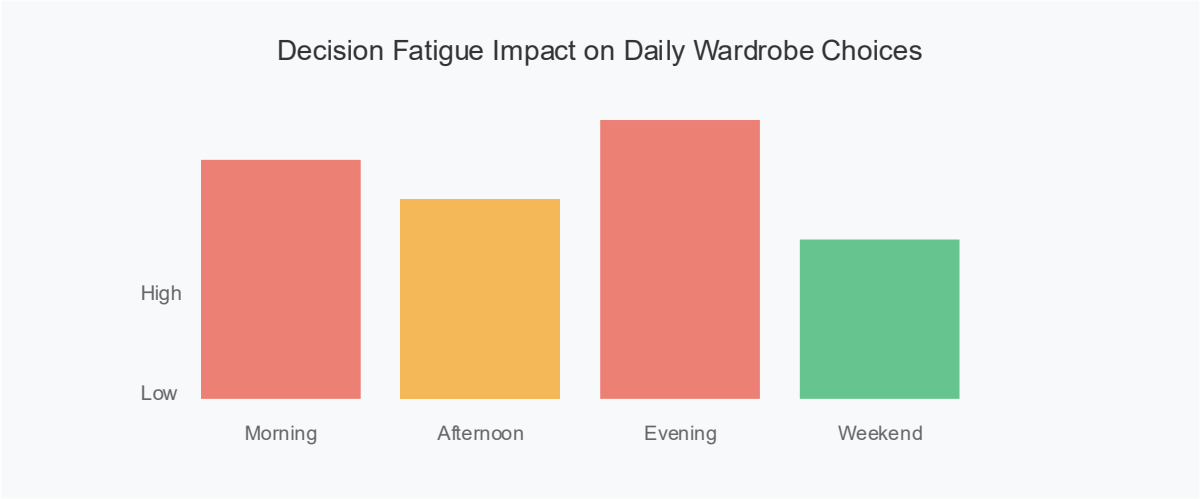


Figure 1. 2 Impact of Decision Fatigue on Wardrobe Selection Throughout the Day

A critical limitation of existing systems is their reliance on weather-based logic designed for temperate, four-season climates. These systems map temperature ranges and weather conditions to appropriate garments, recommending heavy coats for winter and light fabrics for summer [6]. However, in tropical regions like Sri Lanka, where temperatures remain relatively constant throughout the year (25-32°C) and humidity levels frequently exceed 70%, such seasonal logic becomes largely irrelevant. Instead, dressing decisions in tropical climates are governed by micro-climate factors—indoor air-conditioning versus outdoor humidity—and a complex hierarchy of social and cultural events.

Furthermore, existing AI models in fashion recommendation often operate as "Black Boxes," providing suggestions based on visual similarity or collaborative patterns without explaining the stylistic reasoning behind recommendations [7]. Users receive suggestions without understanding why certain items work together, what colour harmony principles apply, or why specific fabrics are appropriate for given conditions. This lack of transparency creates a gap in user trust and educational value, limiting the adoption and effectiveness of AI-driven fashion assistants.

The convergence of several technological advances presents an opportunity to address these limitations. Knowledge Graphs (KGs) have demonstrated remarkable capabilities in capturing complex relationships and enabling explainable reasoning across various domains [8]. Deep learning architectures, particularly Convolutional Neural Networks (CNNs) and Residual Networks (ResNet), have achieved state-of-the-art performance in visual feature extraction for fashion items [9]. Most recently, Diffusion Models have revolutionized generative AI, enabling the creation of photorealistic images from textual descriptions [10].

This research proposes to leverage these technological advances to develop an AI-driven outfit recommendation and matchmaking system that addresses the identified gaps. By integrating semantic knowledge graphs for explainable reasoning, deep learning for visual understanding, and diffusion models for generative capabilities, the proposed system aims to provide a comprehensive solution for intelligent wardrobe management tailored to tropical climate contexts.

1.2 Literature Survey

1.2.1 Fashion Recommendation Systems

Fashion recommendation systems have evolved significantly over the past decade, transitioning from simple collaborative filtering approaches to sophisticated deep learning models. Early systems relied on matrix factorization techniques to predict user preferences based on historical interactions, treating fashion items as generic products without considering their unique visual and semantic characteristics [11].

The introduction of deep learning revolutionized fashion recommendation by enabling direct processing of visual content. CNN-based approaches demonstrated that visual features extracted from clothing images could significantly improve recommendation accuracy compared to traditional methods [12]. Liu et al. introduced the DeepFashion dataset containing over 800,000 labelled images, establishing a benchmark for fashion understanding tasks [13].

Outfit recommendation represents a specialized subset of fashion recommendation focused on compatibility prediction—the task of determining whether multiple items form a cohesive ensemble. Han et al. proposed a Bi-LSTM framework that models the sequential dependencies between outfit items, achieving significant improvements in fill-in-the-blank tasks [14]. More recently, attention mechanisms have been incorporated to identify which specific regions of clothing items contribute most to compatibility judgments [15].

Table 1. 1 Comparison of Existing Fashion Recommendation Systems

System/Approach	Key Technology	Strengths	Limitations
DeepFashion [13]	CNN, Attribute Prediction	Large-scale dataset, comprehensive annotations	No recommendation capability
OutfitNet [15]	CNN, Attention Mechanism	Visual compatibility modelling	Lacks explainability
HFGN [16]	Hierarchical Graph Network	User-outfit-item relationships	No knowledge reasoning
DiFashion [17]	Diffusion Models	Generative outfit creation	Limited personalization

1.2.2 Knowledge Graphs in Fashion

Knowledge Graphs have emerged as a powerful paradigm for representing structured knowledge and enabling reasoning in recommendation systems. Unlike traditional databases that store isolated entities, knowledge graphs capture rich relationships between entities through semantic triples (subject-predicate-object), enabling complex query answering and inference [18].

In the fashion domain, Zhan et al. proposed A3-FKG (Attentive Attribute-Aware Fashion Knowledge Graph), the first comprehensive knowledge graph specifically designed for outfit preference prediction [8]. The graph connects outfits, items, and attributes through multiple relation types, enabling the model to capture fine-grained user preferences. A two-level attention mechanism discriminates individual user's interests toward different outfit attributes while modelling latent multiple tastes.



Figure 1. 3 Knowledge Graph Structure in Fashion Domain

Graph Neural Networks (GNNs) have become the standard approach for learning representations over knowledge graphs. By propagating information through graph structures, GNNs can capture multi-hop relationships and generate embeddings that incorporate neighbourhood context [19]. Recent work has demonstrated that hypergraph extensions of GNNs can better model the complex interactions between multiple items in an outfit [20]. Explainability represents a key advantage of knowledge graph-based approaches. Unlike black-box neural models, knowledge graphs provide explicit reasoning paths that can be traced to explain why particular recommendations are made [21]. This transparency is particularly valuable in fashion recommendation, where users benefit from understanding styling principles that can inform future decisions.

1.2.3 Deep Learning for Visual Feature Extraction

Convolutional Neural Networks have established themselves as the dominant architecture for visual feature extraction in fashion applications. The ResNet (Residual Network) architecture, introduced by He et al., addressed the vanishing gradient problem in deep networks through skip connections that enable training of networks with over 100 layers [22]. ResNet-50, with its 50 layers, has become a standard backbone for fashion image classification and feature extraction.

Recent studies have demonstrated that ResNet achieves superior performance in dress code classification tasks, attaining 83% accuracy compared to 72% for basic CNN architectures [23]. The residual learning framework enables the network to learn fine-grained visual features critical for distinguishing between clothing categories, colours, and styles.

Vision Transformers (ViTs) represent a more recent architectural innovation that applies transformer attention mechanisms directly to image patches [24]. While ViTs have achieved competitive performance on general image classification benchmarks, their application to fashion remains limited by the need for largescale pre-training data. Hybrid approaches that combine CNN-based local feature extraction with transformer-based global attention have shown promise for fashion understanding tasks [25].

1.2.4 Generative AI for Fashion

The advent of Diffusion Models has fundamentally transformed generative AI capabilities. Unlike Generative Adversarial Networks (GANs) that train a generator to fool a discriminator, diffusion models learn to reverse a gradual noising process, enabling stable training and high-quality generation [26]. Stable Diffusion, developed by Rombach et al., operates in a compressed latent space, making high-quality image generation computationally feasible [10].

In fashion applications, diffusion models have enabled novel tasks such as outfit generation from text descriptions and style transfer between garments. Xu et al. introduced DiFashion, a diffusion-based approach for Generative Outfit Recommendation (GOR) that creates entirely new outfit

compositions rather than retrieving existing items [17]. The model achieves parallel generation of multiple fashion images while ensuring compatibility and personalization.

FashionDPO represents a recent advancement that applies Direct Preference Optimization to fine-tune fashion generation models [27]. By incorporating feedback from multiple expert evaluators, the model learns to generate more diverse and aesthetically pleasing outfits. This approach addresses the limitation of supervised learning, which tends to produce homogeneous outputs that closely mirror training examples.

1.2.5 Explainable AI in Recommendation Systems

Explainability has become a critical requirement for AI systems deployed in real-world applications. In recommendation systems, explanations serve multiple purposes: increasing user trust, helping users understand system capabilities, and providing educational value about domain knowledge [28].

Hou et al. proposed SAERS (Semantic Attribute Explainable Recommender System) for fashion recommendation, which generates explanations by highlighting specific semantic attributes (e.g., V-neck collar, short sleeves) that influenced the recommendation [29]. The system projects both users and items into a fine-grained interpretable semantic space, enabling personalized visual explanations.

Techniques for achieving explainability in AI models include feature importance analysis, local explanations through LIME (Local Interpretable Model-agnostic Explanations), and SHAP (SHapley Additive exPlanations) values [30]. Knowledge graph-based approaches offer inherent explainability through explicit reasoning paths, making them particularly suitable for applications where transparency is essential.

1.3 Research Gap

Despite significant advances in fashion recommendation systems, several critical research gaps remain unaddressed in the existing literature. These gaps limit the practical applicability of current approaches and present opportunities for novel contributions.

Climate Context Gap: Existing fashion recommendation systems predominantly adopt weather-based logic designed for temperate, four-season climates. These systems map seasonal variations to appropriate garments without considering the unique challenges of tropical environments. In regions like Sri Lanka, where temperatures remain consistently high and humidity levels are substantial, traditional seasonal logic fails to address the actual factors influencing clothing decisions. The distinction between indoor (air-conditioned) and outdoor (humid) environments becomes more relevant than seasonal changes, yet no existing system adequately models these micro-climate factors [31].

Consumption vs. Utilization Gap: Current e-commerce recommendation systems are optimized for driving new purchases rather than maximizing the utility of existing wardrobes. While collaborative filtering effectively suggests new items similar to past purchases, it does not address the fundamental problem of outfit composition from available inventory. This gap is particularly significant given growing concerns about sustainable fashion and the environmental impact of overconsumption [32].

Explainability Gap: Most AI-driven fashion recommendation systems operate as black boxes, providing suggestions without transparent reasoning. Users receive recommendations based on visual similarity or collaborative patterns but lack understanding of the styling principles underlying these suggestions. This gap limits user trust and prevents the educational value that could be derived from understanding why certain combinations work [33].

Event-Intent Integration Gap: Existing systems typically consider either contextual factors (events, weather) or personal preferences, but rarely integrate both dimensions effectively. The dual requirement of situational appropriateness (what is suitable for the occasion) and personal expression (how the user wants to present themselves) remains unaddressed. This integration is particularly important for complex social contexts such as traditional Sri Lankan weddings or corporate meetings where dress codes intersect with personal style preferences [34].

Generative Personalization Gap: While recent work has explored generative outfit creation, existing approaches lack deep personalization based on learned user style profiles. Generated outfits often appear generic or fail to reflect individual aesthetic preferences. The integration of diffusion models with comprehensive user style understanding remains an open research challenge [35].

Table 1. 2 Research Gaps and Proposed Solutions

Gap Area	Current Status	Proposed Solution
Climate Context	Four-season weather models	Micro-climate fabric performance logic
Consumption Focus	New purchase recommendations	Wardrobe utilization optimization
Explainability	Black-box similarity models	Knowledge graph reasoning paths
Event-Intent	Single-dimension consideration	Dual-layer contextual engine
Generative AI	Generic outfit generation	Style-DNA personalized diffusion

1.4 Research Problem

The rapid expansion of the fashion industry has created a paradox of choice, where individuals face decision fatigue when selecting daily outfits from their existing wardrobes. While e-commerce platforms offer sophisticated recommendation systems, these are primarily designed to drive new consumption rather than optimizing wardrobe utilization. Current Smart Wardrobe solutions either require expensive hardware infrastructure (RFID, sensors) or rely on simplistic weather-based logic inappropriate for tropical climates.

In tropical regions like Sri Lanka, traditional four-season weather models are irrelevant. Instead, dressing decisions are governed by micro-climate factors (indoor AC vs. outdoor humidity) and complex social event hierarchies. Furthermore, existing AI recommendation systems operate as black boxes, providing suggestions without explaining the styling reasoning, creating a gap in user trust and educational value.

The primary research problem addressed by this study is: **How can an AI-driven system leverage semantic knowledge representation, deep learning-based visual understanding, and generative modelling to provide explainable, context-aware outfit recommendations for diverse social events and tropical climates?**

This problem encompasses several sub-problems:

- How can a fashion knowledge graph be constructed and utilized to enable explainable reasoning about outfit compatibility?
- How can micro-climate factors be effectively integrated into clothing recommendations?
- How can diffusion models be personalized to generate outfits aligned with individual style preferences?
- How can event context and personal intent be balanced in recommendation decisions?

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to develop an AI-driven outfit recommendation and matchmaking system that integrates semantic knowledge graphs, deep learning-based visual feature extraction, and diffusion models to provide explainable, contextually-appropriate outfit recommendations optimized for diverse social events and tropical climate conditions.

2.2 Specific Objectives

- **Design and implement a Fashion Knowledge Graph** that captures complex relationships between clothing items, fabrics, colors, events, and styling rules to enable explainable reasoning about outfit compatibility.
- **Develop a Dual-Layer Contextual Engine** that integrates event-driven logic (calendar integration, dress code extraction) with style persona selection to balance situational appropriateness with personal expression.
- **Implement Micro-Climate and Fabric Performance Logic** that analyzes humidity and indoor/outdoor settings to recommend clothing based on material science principles appropriate for tropical environments.
- **Build a CNN-based Visual Feature Extractor** using ResNet architecture to analyze wardrobe items and extract semantic features for compatibility assessment.
- **Create a Generative Vision Studio** using Stable Diffusion models that generates personalized outfit concepts based on learned user Style DNA, extending recommendations beyond existing wardrobe items.
- **Develop a Recommendation Engine** that combines knowledge graph reasoning with collaborative filtering to provide ranked outfit suggestions with transparent explanations.
- **Evaluate system performance** using precision, recall, F1-score, and user satisfaction metrics against baseline approaches.

3. Methodology

3.1 System Overview

The proposed system follows a modular architecture comprising four primary layers: the User Interface Layer, AI Processing Layer, Data Layer, and Recommendation Output Layer. This architecture enables separation of concerns while facilitating integration between components.

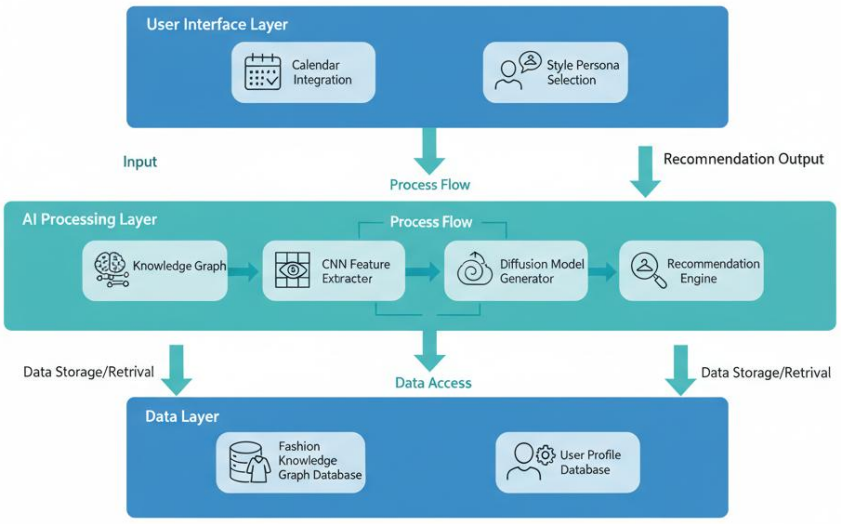


Figure 3. 1 System Architecture Overview

The User Interface Layer provides interaction points for calendar integration, style persona selection, and wardrobe item upload. The AI Processing Layer contains the core intelligence: Knowledge Graph for reasoning, CNN Feature Extractor for visual analysis, Diffusion Model Generator for creative outfit generation, and the Recommendation Engine for ranking and selection. The Data Layer manages the Fashion Knowledge Graph Database and User Profile Database.

3.2 Knowledge Graph Construction

The Fashion Knowledge Graph (FKG) serves as the backbone for explainable reasoning in the system. Unlike traditional relational databases, the knowledge graph captures semantic relationships between fashion entities through a graph structure.

Table 3. 1 Knowledge Graph Entity Types and Relations

Entity Type	Attributes	Example Relations
Clothing Item	Category, Colour, Pattern, Material	pairs_with, complements
Fabric	Breathability, Weight, Texture	suitable_for, avoids
Colour	Hue, Saturation, Temperature	harmonizes_with, contrasts_with
Event	Type, Formality, Cultural Context	requires, permits
Style Rule	Condition, Confidence Score	implies, recommends

The knowledge graph construction process involves three phases:

Schema Definition: Define entity types, attributes, and relation types based on fashion domain expertise. The schema incorporates fashion theory principles including colour theory, fabric science, and dress code conventions.

Knowledge Extraction: Extract entities and relations from multiple sources including fashion literature, style guides, and existing fashion datasets. Natural Language Processing techniques identify styling rules from textual descriptions.

Graph Embedding: Apply Graph Neural Networks to learn vector representations of entities that capture structural relationships. The embeddings enable similarity computation and inference over the graph.

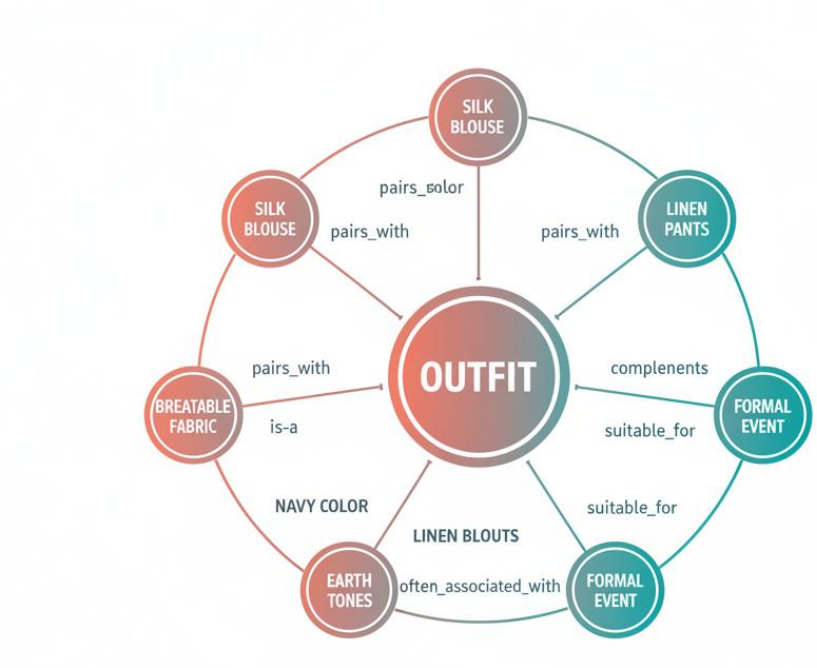


Figure 3. 2 Fashion Knowledge Graph Visualization

3.3 Feature Extraction and Analysis

Visual feature extraction employs a ResNet-50 architecture pre-trained on ImageNet and fine-tuned on fashion-specific datasets. The CNN processes clothing images to extract hierarchical features capturing visual semantics.

Table 3. 2 CNN Architecture Configuration

Layer	Configuration	Output Dimensions
Input	224x224 RGB Image	224 x 224 x 3
Conv1	7x7, 64 filters, stride 2	112 x 112 x 64
MaxPool	3x3, stride 2	56 x 56 x 64
ResBlock 1	3 layers, 64 filters	56 x 56 x 256
ResBlock 2	4 layers, 128 filters	28 x 28 x 512
ResBlock 3	6 layers, 256 filters	14 x 14 x 1024
ResBlock 4	3 layers, 512 filters	7 x 7 x 2048
Global Avg Pool	7x7 pooling	2048

The residual learning framework enables training of deep networks by introducing skip connections that bypass one or more layers:

$$y = F(x, W_i) + x$$

where x is the input, $F(x, W_i)$ represents the residual mapping to be learned, and y is the output. This formulation allows the network to learn identity mappings easily when optimal, while maintaining capacity to learn complex transformations when needed.

Fine-tuning adapts the pre-trained ResNet to fashion-specific features by replacing the final classification layer and training on the DeepFashion dataset. Data augmentation techniques including random rotation, horizontal flipping, and color jittering improve generalization.

3.4 Recommendation Engine

The Dual-Layer Contextual Engine represents a novel contribution of this research, integrating event-driven logic with style persona selection to balance situational appropriateness with personal expression.



Figure 3. 3 Dual-Layer Contextual Engine Architecture

Event-Driven Layer: Calendar integration uses NLP to extract event types and dress codes from calendar entries. A classification model maps event descriptions to formality levels (Casual, Business Casual, Formal, Traditional). The dress code establishes boundaries for appropriate attire.

Style Persona Layer: Within event boundaries, users select a Style Persona (Minimalist, Bold/Statement, Professional, Traditional) that influences color palette selection, silhouette preferences, and accessory choices. The persona acts as a filter on the recommendation space.

Micro-Climate Integration: For tropical climate adaptation, the system analyzes humidity levels and indoor/outdoor settings to prioritize fabrics based on material science principles. Breathable fabrics (cotton, linen) are prioritized for outdoor humidity, while structured layers are suggested for air-conditioned environments.

3.5 Generative Outfit Creation

The Vision Studio component leverages Stable Diffusion models to generate entirely new outfit concepts beyond the user's existing wardrobe. This generative capability serves as a "Virtual Personal Shopper," helping users visualize how new styles or external inspirations might look.



Figure 3. 4 AI-Generated Outfit Concepts from Vision Studio

Style DNA Learning: The system learns a user's "Style DNA" by analyzing their wardrobe items, past outfit selections, and explicit preferences. A personalized embedding captures the user's aesthetic preferences in the latent space of the diffusion model.

Prompt Engineering: Outfit generation uses carefully crafted prompts that incorporate event context, style persona, and learned Style DNA. Example prompt structure:

> *"Professional outfit for corporate meeting, minimalist style, navy blazer with earth-tone trousers, breathable cotton fabric, natural lighting, fashion photography"*

Classifier-Free Guidance: The diffusion sampling process employs classifier-free guidance to balance adherence to the prompt with generation diversity. Higher guidance scales produce more literal interpretations of the prompt, while lower scales allow more creative variation.

3.6 Evaluation Metrics

System evaluation employs both quantitative metrics and qualitative user studies:

Recommendation Accuracy:

- **Precision@K:** Proportion of top-K recommended outfits that users select
- **Recall@K:** Proportion of user-selected outfits that appear in top-K recommendations
- **F1-Score:** Harmonic mean of precision and recall
- **Normalized Discounted Cumulative Gain (NDCG):** Measures ranking quality

Compatibility Prediction:

- **Fill-in-the-Blank (FITB) Accuracy:** Correct selection of missing outfit items
- **Compatibility Score Correlation:** Alignment with human judgments

Explainability Assessment:

- **Explanation Satisfaction:** User ratings of explanation helpfulness
- **Explanation Accuracy:** Expert evaluation of reasoning correctness

User Satisfaction:

- **System Usability Scale (SUS):** Standardized usability questionnaire
- **Net Promoter Score (NPS):** Likelihood to recommend the system
- **Daily Usage Metrics:** Frequency of outfit recommendations requested

3.7 Data Collection and Ethical Considerations

Use of Questionnaire for the Study

A single survey tool shaped this study's data gathering, fitting the work of four researchers each focused on distinct parts. Since every team member handles one segment of the investigation, the questions inside connect directly to each goal area. Responses come together under one structure instead of splitting across several forms. Using one shared method helps keep results aligned while cutting down effort during distribution and replies.

One part of the study looks at how people pick clothes using smart technology. Even though the survey covers many topics tied to the main goals, a few questions zero in on this exact feature. What matters most? How users decide what to wear shows up clearly here. Their answers reveal patterns - what sways choices, which details stand out. Preferences emerge not from assumptions but actual feedback. An automated helper might suggest pairings based on real habits instead of guesses. Feedback shapes whether such tools feel useful or just confusing. Decisions around color, comfort, occasion come through loud and clear. This piece of data anchors the tech side to human behavior.

Questions tied to additional parts of the project play no direct role in shaping this section's findings - still, they support a uniform approach across methods and aid full coverage when gathering information. Focusing on this particular phase, attention stays strictly on answers linked to the AI Outfit Recommendation and Matchmaking feature. These specific inquiries serve to uncover main elements like ease of wear, event type, individual taste, along with less obvious considerations central to building an intelligent suggestion system.

Starting from how people answer certain questions, this work looks for useful patterns to help build a smart tool that recommends outfits. Such a system would rely on personal taste, situational elements, along with unique fashion habits. Instead of generic choices, it responds to the way individuals express their style through answers. The goal emerges not from broad assumptions but from closely examining those replies. Insights grow out of real input rather than theoretical models. With each response shaping part of the process, recommendations become more aligned over time. The questionnaire used for the data collection process is provided in **Appendix A** of this proposal.

3.7.1 Factors influencing the outfit choice.

Table 3. 3 Factors influencing the outfit choice

Factors influence the outfit choice	
Options	**Percentage**
Personal Style	25.9%
Occasion	29.6%
Comfort	37.0%
Weather	3.7%
Fashion Trends	3.7%

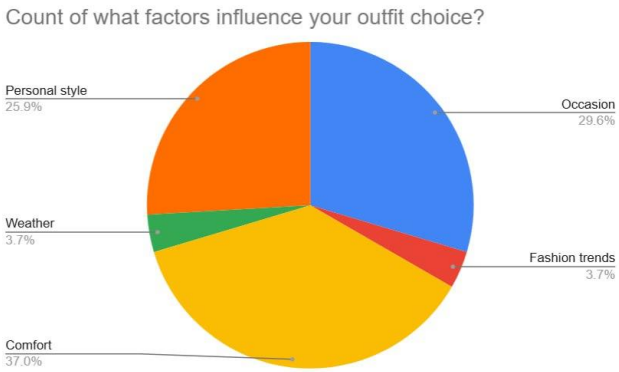


Figure 3. 5 Factors influencing the outfit choice

Looking at the pie chart, it shows how people answered the question about what affects their clothing choices. Comfort stands out as the top reason, making up 37.0% of answers - showing many prefer clothes that feel easy to wear and fit well with everyday life. Although personal style matters too, movement and practicality seem more central for most. Coming next is the setting or event, noted by nearly a third - 29.6% - who adjust attire depending on where they're going. Whether it's work, meeting friends, or dressing formally, the situation plays a clear role. While other aspects exist, these two stand apart in shaping decisions. A quarter of participants point toward self-expression through clothes, showing how attire often mirrors who someone is. While weather matters to some degree, its impact stays limited - only a small fraction cite it as decisive. Fashion waves come and go, yet barely shift choices according to the data collected here. Instead, what stands out is ease paired with purpose: people dress based on where they're headed, how

relaxed they need to feel. Identity threads through selections just as much as convenience does. Context shapes decisions more than external cues like seasonal styles ever do. Comfort isn't just common - it leads alongside personal taste. Responses highlight stability in motives rather than chasing novelty. Knowing this helps shape tools that suggest fits matching real-life habits. Systems tuned to these patterns stand better chances at relevance simply by reflecting actual behavior back to users. Decisions rooted in routine outweigh fleeting influences every time.

3.7.2 Consent to an AI system that suggests outfits from your existing wardrobe?

Table 3. 4 AI outfit Suggestion

AI outfit Suggestion.	
Options	**Percentage**
Yes	59.3%
No	14.8%
Maybe	25.9%

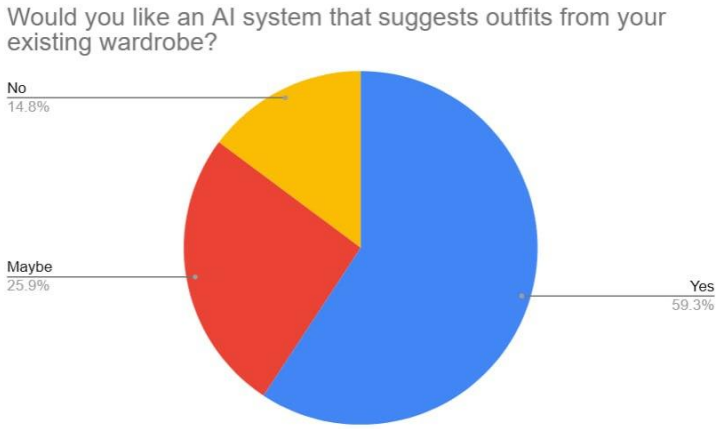


Figure 3. 6 AI outfit Suggestion

This pie chart illustrates the frequency with which individuals struggle to choose an outfit, highlighting a significant prevalence of decision fatigue in daily dressing. A substantial majority of respondents experience this challenge to some degree, with 44.4% reporting they "Sometimes" struggle and 40.7% stating they do so "Very often." Collectively, this indicates that over 85% of the surveyed population faces regular difficulty in outfit selection. In contrast, only a small minority appears confident in their daily choices; just 11.1% "Rarely" struggle, and a mere 3.7% "Never" face this issue. These findings suggest that the process of coordinating attire is a common stressor for most people, potentially pointing toward a need for more streamlined wardrobe management solutions or a better understanding of the psychological factors behind clothing-related decision-making.

3.7.3 Struggle to choose an outfit.

Table 3. 5 Struggle to choose an outfit

Struggle to choose an outfit	
Options	**Percentage**
Very Often	40.7%
Sometimes	44.4%
Rarely	11.1%
Never	3.7%

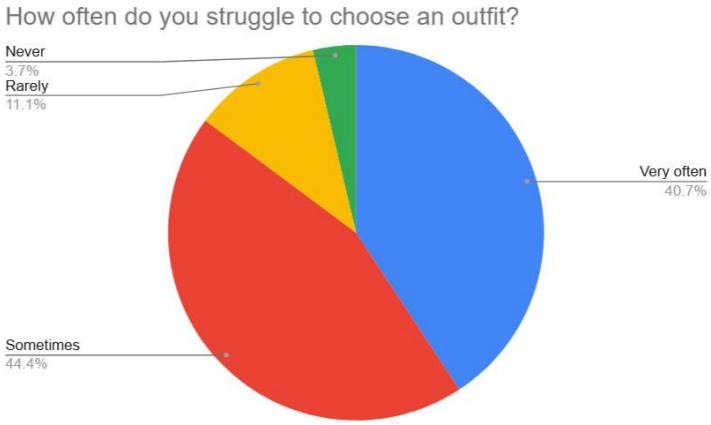


Figure 3. 7 Struggle to choose an outfit

A closer look at the data shows this issue affects numerous people. Nearly half - specifically 44.4% - say they occasionally find it hard to pick an outfit. Because of this pattern, hesitation around clothing choices appears widespread. Reasons might include coordinating pieces, adapting looks to events, or simply not knowing how something fits. While not constant, the challenge surfaces often enough to matter.

Nearly two out of five survey takers say picking clothes takes up too much mental space on regular basis. That figure points to real friction around something many assume should be simple. When those who answered "sometimes" are factored in, the total jumps past eight in ten. Clothing decisions, it turns out, weigh on most people now and then. Such numbers reveal how common the hassle really is across different individuals. What seems minor to some occupies actual daily effort for many others. Frequent hesitation shows the act isn't trivial - it demands energy and attention. More than half cannot just grab things without second thoughts entering mind. Even small choices pile up when repeated each morning without fail. The routine task often drags longer than expected due to uncertainty involved.

Few people sail through wardrobe decisions without pause - just 11.1% say such choices come easily most times. Confidence in dressing leans rare when viewed across the group. At the far end, only 3.7% claim complete ease, standing out as the tiniest cluster within the findings.

Looking at the data, it becomes obvious most individuals struggle somewhat with choosing what to wear. The responses show a clear need for practical approaches that reduce complexity in

picking clothes - options like smart closet organizers or digital styling aids stand out. Instead of guessing each morning, people might rely on these supports to decide faster and feel more certain about their choices, based directly on patterns seen across participant answers

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

The AI-Driven Outfit Recommendation and Matchmaking System is architected as a multi-layered, microservices-based platform that integrates artificial intelligence technologies with fashion domain expertise. The system is designed with modularity, scalability, and maintainability as core architectural principles, enabling independent development, deployment, and scaling of individual components.

The proposed solution comprises three primary architectural layers, each responsible for distinct functional aspects of the system:

Application Services Layer: This layer contains the core business logic implemented as a collection of microservices. Each microservice is responsible for a specific domain function: Wardrobe Management Service, Recommendation Engine Service, Knowledge Graph Service, Generative AI Service, and Calendar Integration Service. Services communicate asynchronously through message queues for event-driven interactions and synchronously through REST for request-response patterns.

AI/ML Processing Layer: The intelligence layer hosts the machine learning models and AI algorithms. This includes the ResNet-50 based visual feature extractor, the Graph Neural Network for knowledge graph embeddings, the Stable Diffusion model for generative outfit creation, and the NLP models for calendar event analysis. These models are deployed as containerized services with GPU acceleration support for inference workloads.

External Integration Layer: This layer manages connections to external services including calendar APIs (Google Calendar, Microsoft Outlook), weather data providers (OpenWeatherMap). All external integrations are abstracted behind service interfaces to enable easy swapping of providers.

Table 4. 1 System Components Summary

Components	Technology Stack	Primary Function
Wardrobe Service	Python/FastAPI	Item upload, classification, metadata management
Recommendation Service	Python/FastAPI	Outfit recommendation generation and ranking
Knowledge Graph Service	Python/Neo4j	Graph queries, reasoning, explanations
Generative AI Service	Python	Outfit image generation using diffusion models
Feature Extractor	Python/TensorFlow	CNN-based visual feature extraction
Calendar Service	Node.js	Calendar integration and event analysis

4.2 System Diagram of Components

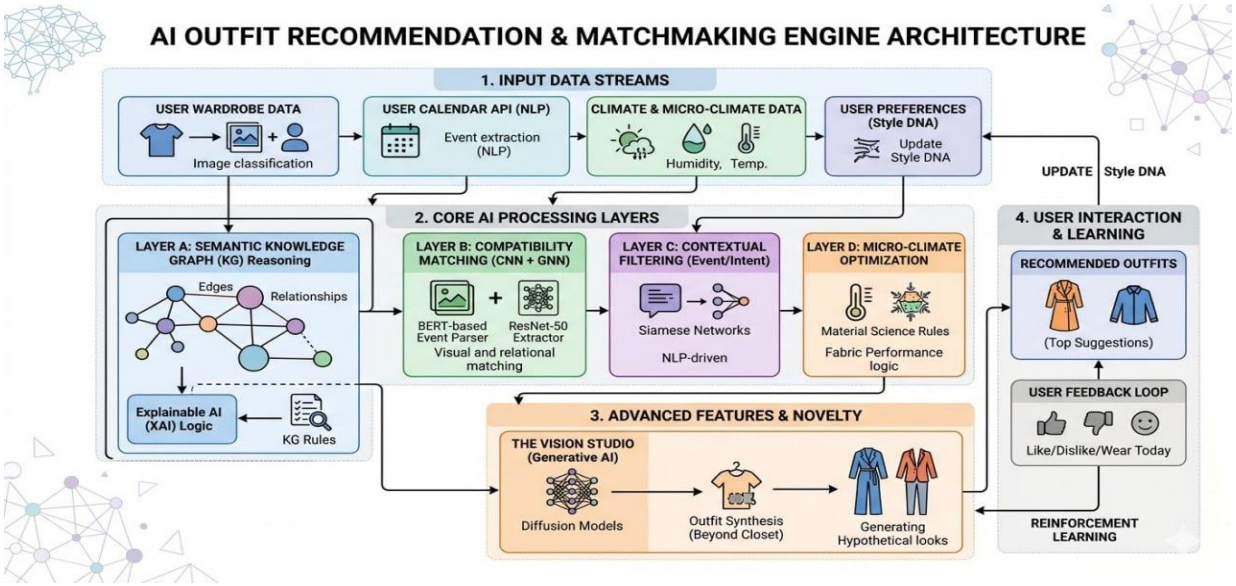


Figure 4. 1 High-Level System Architecture Diagram

4.3 Individual Component and Its Parts

Each component of the system is designed with clear responsibilities, well-defined interfaces, and internal sub-components that work together to deliver the component's functionality.

Wardrobe Management Service

The Wardrobe Management Service provides functionality for users to upload, organize, and manage their clothing items.

Sub-components:

- **Image Upload Handler:** Processes uploaded images, performs validation, compression, and storage to object storage.
- **Item Classifier:** Uses the CNN Feature Extractor to automatically classify items by category, color, pattern, and material.
- **Metadata Manager:** Stores and manages item attributes, purchase information, wear frequency, and user annotations.
- **Search and Filter:** Provides advanced search capabilities across wardrobe items with filters for category, color, season, and occasion.

Recommendation Engine Service

The Recommendation Engine Service is the core intelligence component that generates personalized outfit recommendations.

Sub-components:

- **Context Analyzer:** Analyzes calendar events, weather conditions, and user context to determine recommendation parameters.
- **Compatibility Scorer:** Calculates compatibility scores between clothing items using knowledge graph reasoning and visual feature similarity.
- **Ranking Engine:** Ranks candidate outfits based on compatibility, context appropriateness, and user preferences.

- **Explanation Generator:** Generates natural language explanations for recommendations using knowledge graph paths.

Knowledge Graph Service

The Knowledge Graph Service manages the fashion knowledge graph and provides reasoning capabilities.

Sub-components:

- **Graph Store:** Neo4j database storing entities (items, fabrics, colors, events) and relationships.
- **Query Engine:** Processes Cypher queries for graph traversal and pattern matching.
- **Embedding Generator:** Generates vector embeddings for entities using Graph Neural Networks.
- **Reasoning Engine:** Performs logical inference over graph patterns to derive new relationships.

4.4 Component Interaction

Components interact through well-defined interfaces following RESTful API conventions and asynchronous message passing. The interaction patterns are designed to minimize coupling while ensuring data consistency and system reliability.

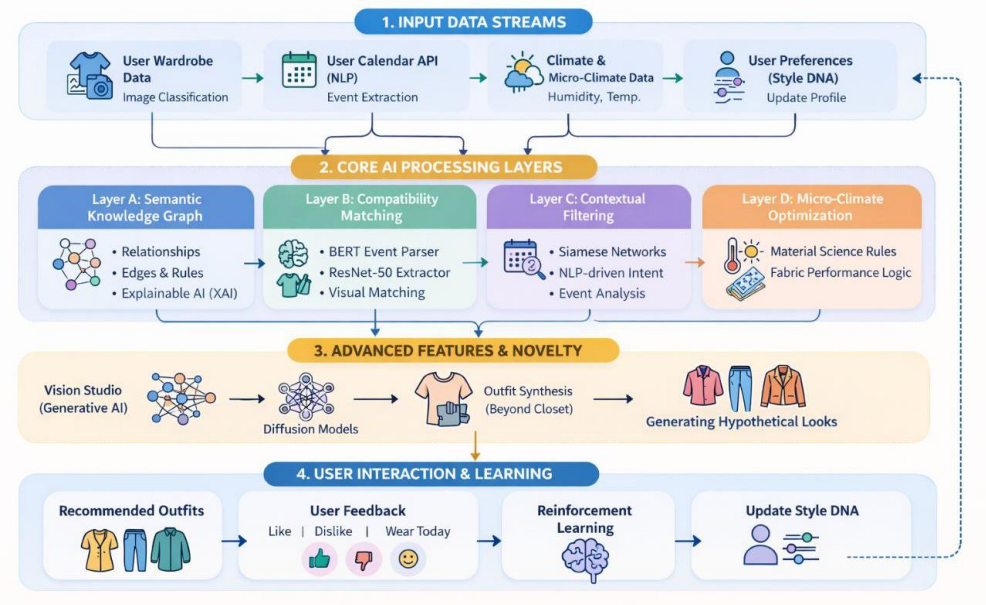


Figure 4. 2 Component Interaction Diagram

Table 4. 2 Component Interaction Matrix

Source Component	Target Component	Interaction Type	Data Exchanged
Wardrobe Service	Feature Extractor	Async Message	Image URL, item ID
Recommendation Service	Knowledge Graph	REST API	Query parameters, context data
Generative AI Service	User Service	REST API	Style DNA embedding

4.5 Data Flow Explanation

Data Flow for Outfit Recommendation:

1. User requests outfit recommendation for a specific date
2. Request flows through API Gateway to Recommendation Service
3. Recommendation Service fetches calendar events from Calendar Service
4. Weather data is retrieved from external weather API
5. User's wardrobe items are queried from Wardrobe Service
6. Knowledge Graph is queried for compatibility rules and styling constraints
7. Recommendation Engine generates candidate outfits
8. Candidates are scored and ranked based on multiple factors
9. Explanations are generated for top recommendations
10. Results are returned to user with images and explanations

Data Flow for Generative Outfit Creation:

1. User provides text prompt or selects inspiration image
2. Request is sent to Generative AI Service
3. Style DNA is retrieved from User Service
4. Prompt Builder constructs optimized prompt combining user input and Style DNA
5. Diffusion Inference Engine generates outfit image
6. Generated image is post-processed and stored
7. Result is returned to user with generation metadata

5. PROJECT REQUIREMENTS

5.1 Functional Requirements

FR1: Wardrobe Management

- The system shall allow users to upload images of clothing items
- The system shall automatically extract visual features and classify items by category, color, and pattern
- The system shall store item metadata including fabric type, purchase date, and wear frequency
- The system shall enable manual editing of item attributes

FR2: Calendar Integration

- The system shall integrate with popular calendar APIs (Google Calendar, Outlook)
- The system shall extract event types and dress codes from calendar entries using NLP
- The system shall classify events into formality levels (Casual, Business Casual, Formal, Traditional)

FR3: Outfit Recommendation

- The system shall generate outfit recommendations based on calendar events
- The system shall allow users to select Style Personas (Minimalist, Bold, Professional, Traditional)
- The system shall filter recommendations by available wardrobe items
- The system shall provide ranked recommendations with compatibility scores

FR4: Explainability

- The system shall provide reasoning explanations for each recommendation
- The system shall highlight specific styling rules applied (color harmony, fabric suitability)

- The system shall display knowledge graph paths supporting recommendations

FR5: Generative Outfit Creation

- The system shall generate new outfit concepts using diffusion models
- The system shall incorporate learned user Style DNA in generation
- The system shall allow text prompts for custom outfit generation

FR6: Micro-Climate Adaptation

- The system shall integrate weather APIs for humidity and temperature data
- The system shall distinguish between indoor (AC) and outdoor environments
- The system shall prioritize fabrics based on climate conditions

5.2 Non-Functional Requirements

NFR1: Performance

- Outfit recommendations shall be generated within 3 seconds
- Image feature extraction shall complete within 2 seconds per item
- The system shall support concurrent users without performance degradation

NFR2: Scalability

- The system shall support wardrobes of up to 500 items per user
- The knowledge graph shall scale to 100,000+ entities
- The system architecture shall allow horizontal scaling

NFR3: Usability

- The user interface shall be responsive and mobile-friendly
- First-time users shall complete onboarding within 10 minutes

- The system shall achieve SUS score above 70

NFR4: Security

- User data shall be encrypted at rest and in transit
- Authentication shall use OAuth 2.0 with multi-factor authentication option
- Wardrobe images shall be stored securely with access controls

NFR5: Reliability

- The system shall maintain 99.5% uptime
- Data backups shall be performed daily
- System failures shall not result in data loss

5.3 User Requirement

5.3.1 Stakeholder Identification

The successful development and deployment of the AI-Driven Outfit Recommendation System require careful consideration of all stakeholders who will interact with, benefit from, or be affected by the system. Stakeholder identification is a critical step in requirements engineering that ensures the system addresses the needs of all relevant parties.

Primary Stakeholders:

End Users (Fashion Consumers): These are the primary beneficiaries of the system, comprising individuals who struggle with daily outfit decisions and seek intelligent wardrobe management assistance. This group includes working professionals who need to dress appropriately for various occasions, students managing limited wardrobes, and fashion-conscious individuals looking to maximize their existing clothing investments. End users require an intuitive interface, accurate recommendations, and explanations that help them understand styling principles.

System Administrators: Technical personnel responsible for deploying, maintaining, and monitoring the system in production environments. Administrators need comprehensive

monitoring dashboards, alerting mechanisms for system anomalies, and tools for user account management. They require access to system logs, performance metrics, and the ability to perform maintenance operations without service disruption.

Secondary Stakeholders:

Fashion Retailers and E-commerce Platforms: Businesses that may integrate with or benefit from the system's recommendations. These stakeholders are interested in API access for outfit recommendations that could drive traffic to their platforms. They require reliable service-level agreements, usage analytics, and integration documentation.

Academic Researchers: Scholars studying fashion recommendation, knowledge graphs, and explainable AI who may use the system as a benchmark or extend its capabilities. This group requires access to published research, open-source components, and documentation of methodologies for reproducibility.

Regulatory Bodies: Organizations concerned with data privacy, algorithmic fairness, and consumer protection. These stakeholders require transparency in data handling practices, evidence of bias mitigation in recommendation algorithms, and compliance with relevant regulations such as GDPR.

5.3.2 User Categories and Needs

Based on the stakeholder analysis, user requirements have been categorized according to the specific needs of each user group:

End User Requirements:

- Easy wardrobe item upload through mobile camera or gallery selection
- Quick outfit recommendations accessible within few seconds of request
- Clear explanations of why outfits are recommended with styling principles
- Ability to explore new styles beyond current wardrobe through Vision Studio
- Climate-appropriate suggestions based on current weather conditions

- Calendar integration for automatic event-aware recommendations
- Style preference learning that improves over time
- Offline access to wardrobe and basic recommendations
- Social sharing capabilities for outfit suggestions
- Privacy controls for wardrobe data and usage analytics

Administrator Requirements:

- User account management capabilities including creation, suspension, and deletion
- System performance monitoring with real-time dashboards
- Knowledge graph maintenance tools for entity and relation updates
- Analytics dashboard showing usage metrics, popular features, and user engagement
- Alerting system for anomalous behavior or system failures
- Backup and recovery tools for database management
- API rate limiting and quota management

5.4 System Requirements

Hardware Requirements:

- Processor (CPU): Intel Core i7 or AMD Ryzen 7 (required for fast AI logic processing).
- Memory (RAM): 16 GB minimum, 32 GB recommended (needed to run the Knowledge Graph and NLP models simultaneously).
- Graphics Card (GPU): NVIDIA RTX 30-series or higher with 6 GB VRAM (essential for training and running Vision/Diffusion models).
- Storage: 512 GB SSD (fast storage is required for processing large fashion datasets and model training).

Software Requirements:

- Operating System: Windows 11, Android 10+ or iOS 13+.
- Programming Language: Python 3.9+ (the primary language for all AI modules).
- Graph Database: Neo4j (to store fashion rules and relationships).
- Vector Database: Pinecone or ChromaDB (to handle image similarity searches).
- PyTorch (for Recommendation models).

Development Tools:

- Git for version control
- Docker for containerization
- Cloud Training: Google Collab Pro

5.5 Expected Test Cases

Table 5. 1 Expected Test Cases

Test ID	Test Case	Expected Output
TC01	Upload clothing image	Image stored, features extracted, item classified
TC02	Request outfit for formal event	Recommendations filtered for formal dress code
TC03	Select Minimalist persona	Neutral colours, simple silhouettes prioritized
TC04	High humidity outdoor setting	Breathable fabrics (cotton, linen) recommended
TC05	Generate new outfit concept	Photorealistic outfit image generated
TC06	View recommendation explanation	Styling rules and knowledge paths displayed
TC07	System under concurrent load	Response time under 3 seconds maintained
TC08	Knowledge graph query	Relevant entities and relations returned

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The commercialization strategy for the AI-Driven Outfit Recommendation System is built upon a thorough understanding of the target market and the development of detailed customer personas that guide product development, marketing, and pricing decisions. The global fashion recommendation market is projected to grow significantly, driven by increasing smartphone penetration, rising fashion consciousness among consumers, and the growing adoption of AI-powered personal styling solutions.

Primary Target Market Segments:

Urban Working Professionals (Ages 25-40): This segment represents the core target market for the initial product. These individuals face daily outfit decisions for professional environments while balancing personal style expression. They typically have moderate to high disposable income, own smartphones, and are comfortable with technology adoption. The segment includes corporate employees, entrepreneurs, and professionals in client-facing roles who understand the importance of appropriate attire. Pain points include time constraints in the morning, uncertainty about dress codes for various occasions, and the desire to maximize wardrobe investments. This segment is estimated at approximately 2.5 million potential users in Sri Lanka and 150 million globally in emerging markets.

University Students and Young Adults (Ages 18-25): This segment represents a high-growth opportunity with strong social media engagement and fashion consciousness. While having lower disposable income, they demonstrate high app engagement and willingness to try new technologies. Their primary needs include budget-friendly wardrobe management, style exploration, and social sharing capabilities. The segment is particularly valuable for viral marketing and user-generated content strategies. The Sri Lankan university student population exceeds 150,000, with a global target market of over 200 million in emerging economies.

Fashion Enthusiasts and Early Adopters (Ages 22-45): This segment comprises individuals who actively follow fashion trends, engage with style content online, and seek innovative solutions for wardrobe management. They are willing to pay premium prices for advanced features and serve

as brand advocates. Their needs include cutting-edge AI features, extensive customization options, and integration with fashion communities. This segment, while smaller, drives product innovation and premium pricing strategies.

Customer Persona - "Professional Priya":

Priya is a 32-year-old marketing manager working in Colombo. She earns LKR 150,000 per month and lives in an urban apartment. Her mornings are rushed as she balances work preparation with family responsibilities. Priya owns approximately 80 clothing items but feels she wears the same outfits repeatedly. She struggles with deciding what to wear for client meetings versus internal team days. Priya uses Instagram for fashion inspiration and has tried several fashion apps but found them focused on shopping rather than wardrobe management. She values convenience, professional appropriateness, and learning styling principles that help her make better fashion decisions independently.

Customer Persona - "Student Sandun":

Sandun is a 22-year-old university student in Peradeniya with a limited clothing budget of LKR 15,000 per month for fashion. He owns about 30 clothing items and wants to maximize their versatility across campus, part-time work, and social events. Sandun is tech-savvy, active on TikTok and Instagram, and values recommendations that help him look stylish without excessive spending. He appreciates features that help him understand why certain combinations work and is likely to share impressive outfit suggestions with friends.

6.2 Value Proposition

The AI-Driven Outfit Recommendation System delivers a unique value proposition that differentiates it from existing solutions in the market. The core value proposition centers on transforming the daily outfit selection experience from a source of stress and decision fatigue into an effortless, educational, and enjoyable activity.

Primary Value Proposition: "Save 10+ days per year while dressing better than ever—your AI stylist understands your wardrobe, your calendar, and the weather to recommend the perfect outfit every morning, with explanations that teach you styling principles for life."

Key Value Drivers:

Time Savings: The system eliminates the average 17 minutes spent daily on outfit decisions, saving users over 100 hours annually. This time can be redirected toward more valuable activities such as exercise, family time, or professional development. For working professionals, this represents tangible productivity improvement.

Wardrobe Optimization: By analyzing existing items and suggesting new combinations, the system helps users maximize the utility of their current wardrobe. Users discover outfits they never considered, effectively expanding their options without new purchases. This addresses both economic and sustainability concerns.

Confidence in Dressing: The explainable AI feature provides users with understanding of why recommendations work, building their personal styling knowledge. Users gain confidence in their fashion choices, knowing they are appropriately dressed for every occasion.

Climate Appropriateness: Unlike competitors focused on temperate climates, the system is specifically designed for tropical environments. Users receive fabric recommendations that account for humidity and indoor/outdoor transitions, ensuring comfort throughout the day.

Creative Exploration: The Vision Studio feature allows users to explore styles beyond their current wardrobe, serving as inspiration for future purchases or simply satisfying creative curiosity. This feature provides value even to users with extensive wardrobes.

6.3 Revenue Model

The commercialization strategy employs a freemium SaaS (Software as a Service) revenue model that balances user acquisition with monetization. This approach allows broad market penetration while generating sustainable revenue from engaged users who derive significant value from premium features.

Free Tier: The free tier provides basic functionality to attract users and demonstrate value. Features include wardrobe management for up to 30 items, daily outfit recommendations, basic weather integration, and limited explanation features. The free tier includes tasteful, non-intrusive

advertising from fashion retailers. This tier serves as a user acquisition channel and conversion funnel for premium subscriptions.

Premium Tier (Style Pro): Priced at LKR 1,500 per month or LKR 12,000 annually (33% discount), the premium tier unlocks unlimited wardrobe items, advanced AI recommendations, full explainability features, calendar integration, Vision Studio access with 50 generations per month, priority support, and ad-free experience. This tier targets working professionals and serious fashion enthusiasts who derive daily value from the service.

Enterprise Tier (Style for Teams): Custom pricing for corporate clients, this tier provides outfit recommendation services for organizations with dress code requirements. Applications include hospitality industry uniform management, corporate image consulting, and retail staff styling guidance. Revenue is generated through annual contracts with per-user licensing.

API Licensing: Third-party developers and fashion retailers can license API access to integrate outfit recommendation capabilities into their platforms. Pricing follows a usage-based model with tiered pricing for different volume levels. This creates an additional revenue stream while expanding the system's reach.

Affiliate Commerce: When users express interest in items similar to generated outfits, the system can provide affiliate links to partner retailers. Revenue is generated through commission on resulting sales. This model aligns incentives by helping users find items they want while generating revenue.

6.4 Cost Estimation

The cost structure for developing and operating the AI-Driven Outfit Recommendation System includes initial development costs, ongoing operational expenses, and marketing investments. Accurate cost estimation is essential for pricing decisions, funding requirements, and profitability projections.

Development Costs (One-Time):

The initial development phase requires investment in personnel, hardware, software licenses, and cloud infrastructure. Development personnel costs include salaries for machine learning engineers, full-stack developers, UI/UX designers, and quality assurance specialists over a 12-month development period. Hardware costs encompass development workstations, GPU servers for model training, and testing devices. Software licenses include development tools, database management systems, and design software. Cloud infrastructure costs during development include compute resources for model training, storage for datasets, and development environment hosting.

Operational Costs (Monthly Recurring):

Ongoing operational costs include cloud infrastructure for production hosting, API services (weather, calendar), customer support, and continuous model improvement. Cloud costs scale with user base and include compute for recommendation generation, storage for user wardrobes, and bandwidth for image serving. API costs depend on usage volume for integrated third-party services. Support costs include personnel for handling user inquiries and technical issues.

Marketing and User Acquisition Costs:

Customer acquisition requires investment in digital marketing, influencer partnerships, app store optimization, and promotional campaigns. Initial launch marketing focuses on building awareness and acquiring the first 10,000 users. Ongoing marketing maintains user growth and engagement through content marketing, social media presence, and referral programs.

6.5 Competitive Advantage

The AI-Driven Outfit Recommendation System establishes sustainable competitive advantages through technological differentiation, domain expertise, and strategic positioning that create barriers to entry for potential competitors.

Technological Differentiation:

The integration of knowledge graphs for explainable reasoning represents a significant technological advantage over competitors relying solely on black-box neural networks. This

differentiation addresses a genuine user need for understanding recommendations, building trust and educational value that competitors cannot easily replicate. The explainability feature also creates network effects as users learn styling principles and become more engaged with the platform.

The tropical climate adaptation provides first-mover advantage in a significant underserved market. While global competitors focus on temperate climate logic, the system's micro-climate fabric performance logic directly addresses the needs of users in South Asia, Southeast Asia, Africa, and Latin America—regions representing over 4 billion people.

The generative Vision Studio feature leveraging state-of-the-art diffusion models positions the system at the forefront of AI fashion technology. This capability requires significant expertise in generative AI and substantial computational resources, creating barriers for smaller competitors.

Data Network Effects:

As the user base grows, the system accumulates valuable data on outfit preferences, style trends, and climate-adaptation patterns. This data improves recommendation quality and enables new features that competitors without similar scale cannot match. The knowledge graph becomes more comprehensive with user interactions, creating a self-reinforcing improvement cycle.

Domain Expertise:

The research foundation of the system provides deep expertise in fashion theory, material science, and cultural dress codes that general-purpose recommendation systems lack. This expertise is embedded in the knowledge graph and styling rules, creating intellectual property that is difficult to reverse-engineer.

Strategic Partnerships:

Partnerships with fashion retailers, textile manufacturers, and climate data providers create ecosystem advantages. Exclusive partnerships for fabric performance data or regional fashion trends provide competitive moats that protect market position.

6.6 Intellectual Property Consideration

Intellectual property protection is essential for maintaining competitive advantage and creating long-term value. The commercialization strategy includes comprehensive IP protection across multiple categories.

Patent Protection:

Novel algorithms and system architectures developed during the research may be eligible for patent protection. Specific innovations warranting patent consideration include the Dual-Layer Contextual Engine architecture, the micro-climate fabric performance scoring algorithm, and the Style DNA embedding methodology for diffusion model personalization. Patent applications will be filed in key markets including Sri Lanka, India, and the United States.

Copyright Protection:

Software code, user interface designs, documentation, and original content are protected under copyright law. All code will include appropriate copyright notices, and access to source code will be controlled through licensing agreements.

Trade Secrets:

Certain proprietary knowledge including detailed knowledge graph structures, training data curation methods, and model optimization techniques will be maintained as trade secrets. Employee and contractor agreements will include appropriate non-disclosure and non-compete clauses.

Trademark Protection:

The product name, logo, and associated branding elements will be trademarked to prevent market confusion and protect brand value. Trademark registration will cover relevant classes for software services and fashion consulting.

Open-Source Strategy:

Selected non-core components may be released as open source to build community engagement and establish technical credibility. The open-source strategy will be carefully managed to protect core IP while benefiting from community contributions.

7. BUDGET AND JUSTIFICATION

The budget for the AI-Driven Outfit Recommendation System project encompasses all costs required for successful development, deployment, and initial operation of the system. This section provides detailed cost breakdowns with justifications for each expense category, ensuring transparency and accountability in resource allocation.

7.1 Hardware Costs

Hardware investments are essential for development, training, and deployment of the AI-driven system. The hardware requirements are categorized into development workstations, training infrastructure, and production servers.

Table 7. 1 Hardware Cost Breakdown

Item	Quantity	Unit Cost (LKR)	Total Cost (LKR)
Development Laptop (Intel i7, 32GB RAM, RTX 3060)	1	Owned	Owned
GPU Server (RTX 4090, 64GB RAM) - Training	1	15,000	15,000
External SSD Storage (4TB)	2	10,000	20,000
Test Mobile Devices (Android & iOS)	2	Owned	Owned
Backup Power Supply (UPS)	1	75,000	75,000
Networking Equipment	1	50,000	50,000
Hardware Subtotal			**160,000**

7.2 Software Licenses

Software licenses are required for development tools, design software, and specialized applications. Where possible, open-source alternatives are utilized to minimize costs while maintaining functionality.

Table 7. 2 Software License Costs

Software	License Type	Duration	Total Cost (LKR)
Adobe Creative Cloud	Student License	1 Year	Free
Microsoft Office 365	Student License	1 Year	Free
Neo4j Enterprise (Evaluation)	Student License	1 Year	5,000
GitHub Pro	Student Pack	1 Year	6,000
Individual License	Student License	1 Year	10,000
Software Subtotal			**21,000**

7.3 Cloud Services

Cloud services are essential for hosting, model training, and API integrations. The budget allocates resources for development, testing, and initial production deployment.

Table 7. 3 Cloud Services Cost Estimation

Service	Provider	Monthly Cost (LKR)	Annual Cost (LKR)
Compute Instances (GPU for training)	AWS/GCP	25,000	300,000
Database Hosting (MongoDB)	AWS RDS	Free	Free
CDN for Image Delivery	Cloudflare	5,000	60,000
Weather API Subscription	OpenWeatherMap	4,000	48,000
Domain & SSL Certificate	Various	2,000	24,000
Cloud Services Subtotal (Annual)			**432,000**

7.4 Human Cost Efforts

The human effort cost represents the value of researcher time invested in the project. While this is an academic research project, estimating effort costs provides context for the total investment and enables comparison with commercial development projects.

Table 7. 4 (Continued) Human Effort Cost Estimation

Role	Hours/Week	Duration (Weeks)	Rate (LKR/hr)	Total Cost (LKR)
Researcher (Primary Developer)	50	24	0	0
Supervisor Consultation	2	24	0	0
External Review & Testing	1	3	1,000	3,000
Human Effort Subtotal				**3,000**

7.5 Justifications for Each Expense

Hardware Justification:

The development laptop with dedicated GPU (RTX 3060) is essential for local development and testing of deep learning models. While cloud GPUs are available for training, local development capability significantly accelerates the iteration cycle. The GPU server with RTX 4090 is required for training the diffusion models and large-scale knowledge graph embeddings, tasks that require substantial VRAM and compute power. External SSD storage provides reliable backup for datasets and model checkpoints, critical for protecting work against hardware failures. Test mobile devices ensure the application works correctly across different platforms and screen sizes.

Software License Justification:

Adobe Creative Cloud enables professional design work for UI/UX elements and marketing materials. The use of student licenses for Microsoft Office, GitHub Pro, and JetBrains products represents significant cost savings while maintaining full functionality. Neo4j's academic license provides enterprise-grade graph database capabilities at no cost for research purposes.

Cloud Services Justification:

GPU compute instances are necessary for training the ResNet feature extractor and Stable Diffusion models, tasks that require days of continuous GPU computation. Managed database services reduce operational overhead and provide automatic backups. CDN ensures fast image loading for users regardless of geographic location. Weather API integration is essential for the micro-climate adaptation feature, a key differentiator of the system.

Human Effort Justification:

The researcher time allocation of 50 hours per week over 24 weeks reflects the substantial effort required for a project of this scope. This includes time for literature review, system design, implementation, testing, and documentation. The supervisor consultation rate reflects the value of expert guidance in research methodology and technical architecture. External review and testing ensures the system meets quality standards and user needs before deployment.

Table 7. 5 Total Project Budget Summary

Category	Amount (LKR)
Hardware Costs	160,000
Software Licenses	21,000
Cloud Services	432,000
Human Effort	3,000
Total Project Budget	**616,000**

8. WORK BREAKDOWN STRUCTURE (WBS)

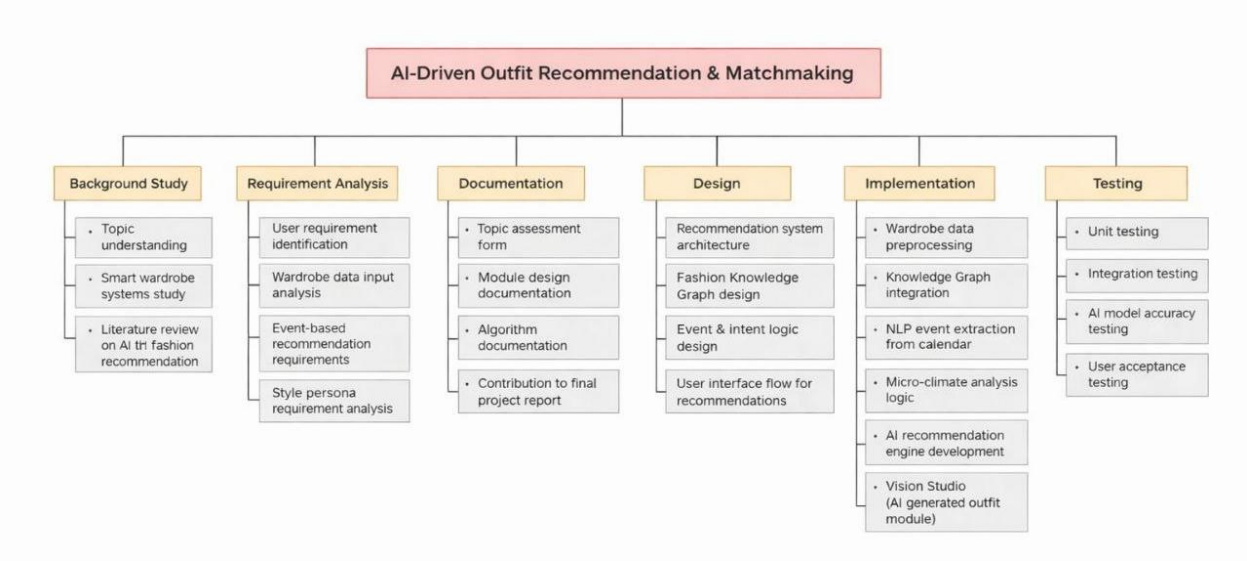


Figure 8. 1 Work Breakdown Structure

9. Gantt Chart

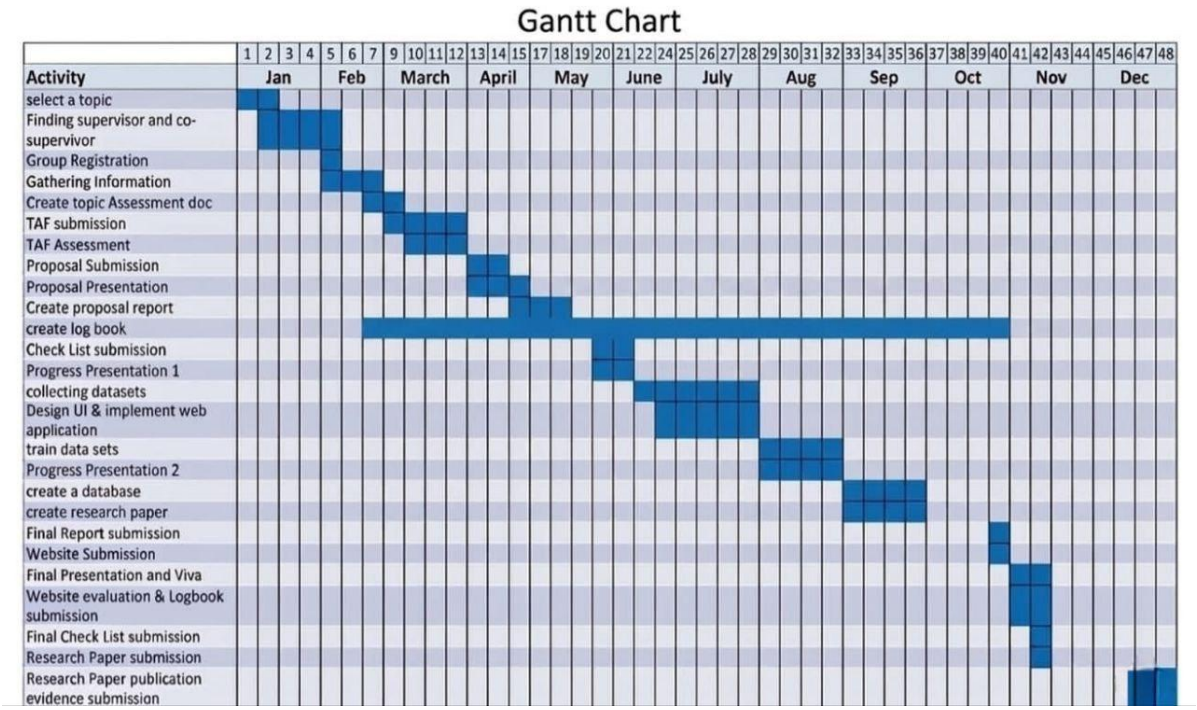


Figure 9. 1 Project Timeline Gantt Chart

10. DESCRIPTION OF PERSONNEL AND FACILITIES

10.1 Personnel

This component of the research is carried out independently by Nasardeen Mohammed Fathima Nisfa, Student ID- IT22909428, an undergraduate student in the BSc. (Hons) in Information Technology specializing in Information Technology at the Sri Lanka Institute of Information Technology.

The researcher is responsible for all aspects of the AI-Driven Outfit Recommendation and Matchmaking System development.

Researcher Responsibilities:

- **Knowledge Graph Development:** Design and implement the Fashion Knowledge Graph schema, extract entities and relations from fashion literature, and implement graph embedding algorithms using Graph Neural Networks.
- **Deep Learning Model Implementation:** Fine-tune ResNet-50 architecture for fashion feature extraction, implement CNN-based visual analysis pipeline, and optimize model performance through hyperparameter tuning.
- **Recommendation Engine Development:** Implement the Dual-Layer Contextual Engine integrating event-driven logic with style persona selection, develop micro-climate adaptation algorithms, and build the recommendation scoring function.
- **Generative AI Integration:** Fine-tune Stable Diffusion models for outfit generation, implement Style DNA learning algorithms, and develop the Vision Studio interface.
- **System Integration:** Develop REST APIs for component communication, implement calendar integration, and build the user interface.
- **Evaluation and Testing:** Design evaluation protocols, conduct user studies, analyze results, and iterate on system improvements.

Supervision: The project is supervised by an academic supervisor from the Department of Information Technology who provides guidance on research methodology, technical architecture, and evaluation design.

10.2 Facilities

Computing Resources:

- Personal laptop with Intel Core i7 processor, 16GB RAM, AMD Ryzen 5000 Series and AMD Radeon Graphics GPU for development and testing.
- NVIDIA RTX 30 Series with 6GB RAM for training and running Vision/Diffusion Module.
- Access to SLIIT computing labs with high-performance workstations.
- Cloud computing credits to MongoDB and Google Collab Pro for model training and deployment.

Software and Tools:

- **Development Environment:** Visual Studio Code
- **Deep Learning Frameworks:** PyTorch 2.0
- **Knowledge Graph:** Neo4j Community Edition, NetworkX
- **Database:** MongoDB
- **Version Control:** Git, GitHub
- **Project Management:** Trello
- **Documentation:** Microsoft Office, LaTex

Dataset Access:

- DeepFashion dataset for fashion image classification
- Polyvore dataset for outfit compatibility

- Custom wardrobe dataset collected through user studies

Internet and Connectivity:

- High-speed internet connection for cloud services and dataset downloads

Access to academic databases (IEEE Xplore, ACM Digital Library) through SLIIT library

APPENDICES

Appendix A – Questionnaire

Section 1: Basic Information

1. Name (Optional)

2. Gender

- Male
- Female
- Non-binary
- Prefer not to say

3. What is your age group?

- Under 18
- 18 – 25
- 26 – 35
- 36 – 45
- Above 45

4. How often do you buy clothes?

- Once a month
- Every few months
- Once or twice a year
- Rarely

5. Approximately how many clothing items do you own?

- Less than 20
- 20 – 50
- 50 – 100
- More than 100

Section 2: Wardrobe Organization

6. Do you find it difficult to organize your wardrobe or remember what clothes you own?

- Yes
- Sometimes
- No

7. Do you currently use any digital wardrobe or outfit planning application?

- Yes
- No

8. How useful would an application be that automatically organizes your wardrobe by clothing type, color, and usage frequency?

- Very useful
- Useful
- Neutral
- Not useful

Section 3: Outfit Recommendation / Matchmaking

9. How often do you struggle to choose an outfit from your existing wardrobe?

- Very often
- Sometimes
- Rarely
- Never

10. Would you like an AI system that suggests outfit combinations from your existing clothes?

- Yes
- Maybe
- No

11. What factors influence your outfit choice the most? (Multiple choice)

- Weather conditions
- Occasion / event
- Personal style
- Comfort
- Social trends

Section 4: Virtual Try-On / Avatar System

12. Have you ever used a virtual try-on application?

- Yes
- No

13. What type of virtual try-on systems have you used before?

- Glasses try-on
- Makeup try-on
- Clothing try-on
- None

14. Would you like to preview clothing outfits on a personalized 3D avatar that resembles you?

- Yes
- Maybe
- No

15. How important is it for the virtual avatar to move naturally while wearing clothes?

- Very important
- Important
- Neutral
- Not important

Section 5: Clothing Image Classification

16. Would you prefer a system that automatically detects clothing types when you upload a photo?

- Yes
- Maybe
- No

17. How helpful would automatic clothing classification be for managing your wardrobe?

- Very helpful
- Helpful
- Neutral
- Not helpful

Section 6: Sustainability & Usage

18. Do you often forget about clothes that you already own?

- Yes
- Sometimes
- No

19. Would you like an application that tracks how frequently your clothes are worn?

- Yes
- Maybe
- No

20. Which features would you like most in a smart wardrobe application? (Multiple choice)

- Automatic clothing classification
- Outfit recommendations
- AR outfit visualization
- Personalized 3D avatar
- Wardrobe usage tracking

21. Would you be willing to use a mobile application that combines AI wardrobe management and AR outfit visualization?

- Yes
- Maybe
- No

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